

DWEP+ Executive Innovation Report

DWEP+ is a smart urban-industrial water exchange framework designed to address industrial freshwater scarcity and untreated municipal wastewater discharge in Durgapur.

1. Problem Statement

Durgapur industries face severe water shortages while municipal wastewater continues to be discharged into the Damodar River. This creates both environmental and economic challenges.

2. DWEP+ Solution Framework

Component	Purpose
IoT Monitoring	Real-time quality sensing
AI Matching	Dynamic industrial allocation
Blockchain Ledger	Transparent smart contracts
Water Exchange	Municipal-industrial reuse system

3. Economic Feasibility

Metric	Value
Unit Revenue	₹45 / KL
Unit Cost	₹25 / KL
Profit Margin	44%
Projected Revenue	₹150 Cr / Year

4. Sustainability Impact

DWEP+ supports circular water economy principles, reduces untreated wastewater discharge, and decreases industrial groundwater dependency.

5. Implementation Roadmap

Phase 0: Pilot Cluster → Phase 1: Municipal Integration → Phase 2: Regional Expansion → Phase 3: National Scale-Up

6. Conclusion

DWEP+ transforms wastewater into a reusable industrial resource through AI, IoT, and smart-city infrastructure integration.